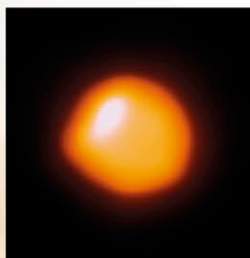


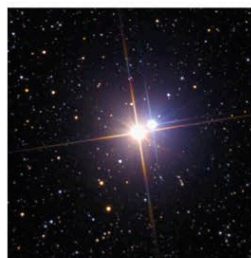
STARS AND BROWN DWARFS

The universe's light comes mainly from stars—hot balls of gas that generate energy through nuclear fusion in their cores. Stars form from the condensation of clumps of gas and dust in nebulae and sometimes occur in pairs or clusters. Depending on their initial mass, stars vary in color, surface temperature, brightness, and life span. The most massive stars, known as giants and supergiants, are the hottest and brightest, but last for only a few million years. Low-mass stars (the most numerous) are small, dim, red, and may live for billions of years; they are called red dwarfs. Smaller still are brown dwarfs. These are failed stars, not massive or hot enough to sustain the type of fusion that occurs in stars, and emit only a dim glow. Brown dwarfs may account for much of the ordinary matter in the universe.



SUPERGIANT STAR

The supergiant Betelgeuse is so vast, it can be seen as a blob in large modern telescopes, though it is some 650 light-years away.

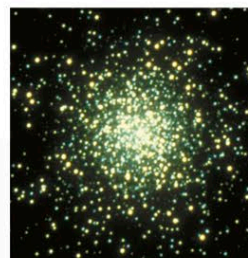


DOUBLE STAR

Albireo is a binary, or double star, consisting of a bright yellow-orange main star and a dimmer, bluish companion.

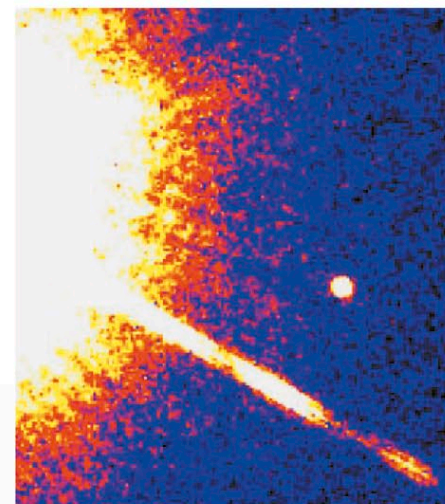
BROWN DWARF

The dot to the right of center in this picture is a brown dwarf called Gliese 229b. The bigger, brighter object is the red dwarf star Gliese 229, around which it orbits.



GLOBULAR CLUSTER

Star clusters such as M3, above, are ancient objects that orbit galaxies. M3 has about half a million stars.



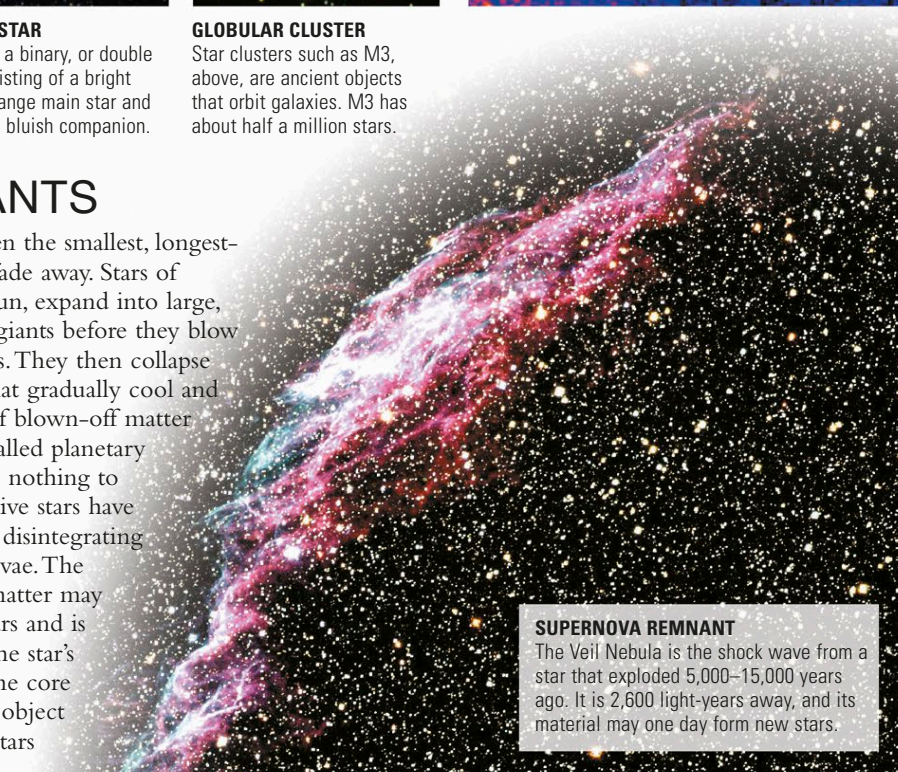
PLANETARY NEBULA

This glowing cloud of gas, called NGC 6751, was ejected several thousand years ago from the hot white dwarf star visible at its center.

STAR REMNANTS

Stars do not last forever. Even the smallest, longest-lived red dwarfs eventually fade away. Stars of medium mass, such as the Sun, expand into large, low-density stars called red giants before they blow off most of their outer layers. They then collapse to form white dwarf stars that gradually cool and fade. The expanding shells of blown-off matter surrounding such stars are called planetary nebulae (although they have nothing to do with planets). More massive stars have even more spectacular ends, disintegrating in explosions called supernovae. The expanding shell of ejected matter may be seen for thousands of years and is

called a supernova remnant. Not all of the star's material is blown off, however. Part of the core collapses to a compact, extremely dense object called a neutron star. The most massive stars of all collapse to black holes (see p.26).



SUPERNOVA REMNANT

The Veil Nebula is the shock wave from a star that exploded 5,000–15,000 years ago. It is 2,600 light-years away, and its material may one day form new stars.

PLANETS AND SMALLER BODIES

The solar system (our own star, the Sun, and everything that orbits it) is thought to have formed from dust and gas that condensed into a spinning disk called a protoplanetary disk. The central material became the Sun, while the outer matter formed planets and other small, cold objects. A planet is a sphere orbiting a star and, unlike a brown dwarf, produces no nuclear fusion. As thousands of planets have now been detected orbiting stars elsewhere in our galaxy (see pp.296–299), it now seems almost certain that planets are common throughout the universe. In the solar system, the planets are either gas giants, such as Jupiter, or smaller, rocky bodies, such as Earth and Mars. Still smaller objects fall into five categories. Moons are objects that orbit planets or asteroids. Asteroids are rocky bodies of about 150 ft (50 m) to 600 miles (1,000 km) across. Comets are chunks of ice and rock a few miles in diameter that mostly orbit in the far reaches of the solar system. Ice dwarfs are similar but are up to a few hundred miles across. Meteoroids are the remains of shattered asteroids or dust from comets.



IO



EUROPA



GANYMEDE



CALLISTO



PLANET EARTH

Our home planet seems unusual in having surface water and supporting life. We do not know how rare this is in the universe.

GALILEAN MOONS

Other than Earth's own Moon, these four large moons orbiting the planet Jupiter were the first ever discovered, by Galileo Galilei in 1610.



COMET IKEYA-ZHANG

A few comets travel in orbits that bring them close to the Sun. Frozen chemicals in the comet then vaporize to produce a glowing coma (head) and long tails of dust and gas. This bright comet was visible in 2002.